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Rinku Shah

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Current Role and Affiliation

Assistant Professor, Department of Computer Science & Engineering, IIIT-Delhi

Research Interests

As a networked systems researcher, I build flexible and scalable systems that solve real-world problems in the intersection of networks, security, and AI. My broad areas of interest include:

- Networks/Systems for AI
- Software-defined Networking (SDN)
- Programmable Data Planes

Education

2013 – 2021 **Indian Institute of Technology Bombay**, *Ph.D., Computer Science & Engineering*, Defense date: February 7, 2021; *GPA – 9.08*.
Advisors: Prof. Mythili Vutukuru and Prof. Purushottam Kulkarni
Thesis title: Hierarchical Control Plane Designs to Scale SDN Applications

2007 – 2010 **Ramrao Adik Institute of Technology**, *Masters, Computer Engineering*, Mumbai University, *Percentage – 75%*.
First Class with Distinction

1998 – 2002 **Shivajirao.S.Jondhale College of Engineering**, *Bachelors, Computer Engineering*, Mumbai University, *Percentage – 61%*.
First Class

Professional Experience

July 2021 – present **Indraprastha Institute of Information Technology Delhi**, *New Delhi, IN*, Assistant Professor, Department of CSE.

Nov 2020 – May 2021 **IIT Bombay**, *Project Research Scientist (Huawei project)*, Mumbai, IN.
Principal Investigators: Umesh Bellur, Purushottam Kulkarni
Project Title: High Performance Distributed Object Store

Fall 2019 **VMware**, *Research Intern*, Pune, IN.
Mentors: Sairam Veeraswamy, T. Shridhar, Raja Kommula, Rohan Gandhi, Abhijit Sharma
Project Title: Capacity-aware Load Balancing for Network Functions

2010 – 2016 **Vidyalankar Institute of Technology**, *Assistant Professor*, Mumbai, IN.

2003 – 2010 **Vidyalankar Institute of Technology**, *Lecturer*, Mumbai, IN.

Spring 2003 **St. Francis Institute of Technology**, *Lecturer*, Mumbai, IN.

Publications

- Palak Mishra, Rajat Bhardwaj, Yash Verma, Ramanjeet Singh, Rinku Shah. Enabling Hardware-Software Co-Design for Large-Scale AI Systems via Fine-Grained GPU Memory Tracing.
  In Proceedings of the 1st Workshop on ACM MemNetAI 2026, co-located with SIGCOMM 2026.
- Aditya Peer, Rohan, Siddhant, Neeraj, Amandeep, Praveen Tammana, Sumit Darak, Rinku Shah. Designing size-aware cryptographic primitives for FPGA-based accelerators.
  In Proceedings of the Proceedings of the ACM on Networking, Volume 4, Issue CoNEXT1 (March 2026), 2026.
- Sumit Kumar, Arjun, Naman, Ramanjeet, Meet, Praveen, Satananda, Abed, Rinku Shah. Simulating LLM training workloads for heterogeneous compute and network infrastructure.
  In Proceedings of the 2nd Workshop on Networks for AI Computing, 2025, colocated with ACM SIGCOMM 2025.
- Ranjitha K, Medha, Stavan, Karuturi, Sri Hari, Dheekshitha, Rinku Shah, Praveen Tammana. Securing In-Network Traffic Control Systems with P4Auth.
  In Proceedings of the 55th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2025.
- Ranjitha K, Karuturi Havya Sree, Devansh Garg, Stavan Christian, Dheekshitha Bheemanath, Rinku Shah, Praveen Tammana. Detecting Manipulation to Table Rules in the Programmable Data Planes.
  In Proceedings of the IFIP/IEEE Networking Conference (IFIP Networking), 2025.
- Siddharth Nayak, Vishesh Rangwani, Kartikay Dubey, Rajorshi Mondal, Tushar Gupta, Rinku Shah. Detecting Manipulation to Table Rules in the Programmable Data Planes.
  In Proceedings of the 20th International Conference on Emerging Networking EXperiments and Technologies (CoNEXT), 2024.
- Prashanth P S, Ranjitha K, Ankit Sharma, Arjun Temura, Rinku Shah, Praveen Tammana. DL^3: Adaptive Load Balancing for Latency-critical Edge Cloud Applications.
  In Proceedings of the International Conference on Network and Service Management (CNSM), 2024.
- Vedant Bothra, Aditya Peer, Vijay Kumar Singh, Mukulika Maity, Rinku Shah. CoDel-ACT: Realizing CoDel AQM for Programmable Switch ASIC.
  In Proceedings of the IFIP/IEEE Networking Conference (IFIP Networking), 2024.
- Harish S A, K Shiv Kumar, Anibrata Majee, Amogh Bedarakota, Praveen Tammana, Pravein Kannan, Rinku Shah. In-Network Probabilistic Monitoring Primitives under the Influence of Adversarial Network Inputs.
  In Proceedings of the ACM Asia-Pacific Workshop on Networking (APNet), 2023.
- Lasani Hussain, Mayank Rawat, Neeraj Kumar Yadav, Sumit Darak, Praveen Tammana, Rinku Shah. Microservice-based in-network security framework for FPGA NICs.
  In Proceedings of the IEEE/ACM CCGRIDW 2023.
- Abhik Bose, Shailendra Keshav Kirtikar, Shivaji Chirumamilla, Rinku Shah, Mythili Vutukuru. AccelUPF: Accelerating the 5G user plane using programmable hardware.
  In Proceedings of the ACM SIGCOMM Symposium on SDN Research (SOSR), 2022
- Shaguftha Kottur, Krishna Kadiyala, Praveen Tammana, Rinku Shah. Implementing ChaCha Based Crypto Primitives on Programmable SmartNICs.
  In Proceedings of the ACM SIGCOMM Workshop on FFSPIN, 2022.
- S. VenkataKeerthy, Yashas Andaluri, Sayan Dey, Rinku Shah, Praveen Tammana, and Ramakrishna Upadrasta.Packet Processing Algorithm Identification using Program Embeddings.
  In Proceedings of the ACM 5th Asia-Pacific Workshop on Networking (APNet), 2022.

- ✻ Abhik Bose, Diptyaroop Maji, Prateek Agarwal, Nilesh Unhale, Rinku Shah, Mythili Vutukuru. Leveraging Programmable Dataplanes for a High Performance 5G User Plane Function. In Proceedings of the ACM 5th Asia-Pacific Workshop on Networking (APNet), 2021.

Selected publications (2020 and older).

- ✻ Rinku Shah, Vikas Kumar, Mythili Vutukuru, Purushottam Kulkarni. TurboEPC: Leveraging data plane programmability to accelerate the mobile packet core. In Proceedings of the ACM SIGCOMM Symposium on SDN Research (SOSR), 2020.
- ✻ Rinku Shah, Mythili Vutukuru, Purushottam Kulkarni. Cuttlefish: Hierarchical SDN Controllers with Adaptive Offload. In Proceedings of the IEEE 26th International Conference on Network Protocols (ICNP), 2018.
- ✻ Rinku Shah, Aniket Shirke, Akash Trehan, Mythili Vutukuru, Purushottam Kulkarni. pcube: Primitives for network data plane programming. In Proceedings of the IEEE 26th International Conference on Network Protocols (ICNP), 2018.
- ✻ Rinku Shah, Mythili Vutukuru, Purushottam Kulkarni. Devolve-Redeem: Hierarchical SDN Controllers with Adaptive Offloading. In Proceedings of the ACM First Asia-Pacific Workshop on Networking (APNet), 2017.

The complete list of publications is available at my Google Scholar page

Research Grants

2026–2030 Anusandhan National Research Foundation (ANRF)

Project title: INDRA-OT: Integrated Network Defense Response and Analysis for OT and SCADA Networks
Grant support (as Co-PI): Rs. 233.37 lakhs; PI: Sambuddho Chakravarty

2025–2028 Marvell India Private Limited

Project title: RDMA NIC simulation tools for distributed ML workloads
Grant support (PI): Rs. 86 lakhs

2024–2026 IBM OSCP Faculty Fellowship Award
Grant support: approx Rs. 2.5 lakhs

Project title: Programmable Observability

2023–2026 Cybersecurity and Cybersecurity for Cyber-Physical Systems Innovation Hub (C3iHub)

Project title: Programmable cryptosystem for 5G telecommunication networks
Grant support (PI): Rs. 32 lakhs

2023–2026 Cybersecurity and Cybersecurity for Cyber-Physical Systems Innovation Hub (C3iHub)

Project title: Siegebreaker II: End-to-middle Covert Internet Communication Using Programmable Network Hardware
Grant support (as Co-PI): Rs. 28 lakhs; PI: Sambuddho Chakravarty

2023-2024 NIDHI PRAYAS Program (Cohort 3)

Project title: FastCrypt — a smart security hardware for data centers
Grant support (PI): Rs. 7 lakhs

2023-2027 DST PURSE grant

Project title: Smart Secure Connected Autonomous Systems Research
Grant support (PI group member – Num PIs=21): Rs. 6.51 crores

Grants procured to conduct professional activities

Dec 2023 ACM India Winter School grant
Grant support: approx Rs. 6 lakhs

For Winter School on Full-stack Networking

Dec 2023 NMICPS TiHAN Foundation, IIT Hyderabad
Grant support: Rs. 2 lakhs
For Winter School on Full-stack Networking

Professional Activities (recent)

Core Organising Committee Member

2026 Workshop Co-Chair, Memory-Semantic Networking for AI-Scale Systems (MemNetAI), colocated with ACM SIGCOMM 2026
2026 General Co-Chair, Indian Symposium on Computer Systems (IndoSys)
2026 Workshop Co-Chair, Networking and Fabric Architectures for AI Systems (NetFabAI), colocated with IEEE/IFIP Networking 2026
2024 Local Chair, Building a Secure & Empowered Cyberspace (BuildSec)
2023 Academic & Local Co-ordinator, ACM India Winter school on Full-stack Networking

Technical Program Committee member

2026 ACM/IFIP International Middleware Conference
2023, 2024 COMSNETS
2022 ACM SIGMETRICS PerfNA Workshop
2022 ACM SIGCOMM Symposium on SDN Research (SOSR)
2023, 2024 ACM India ARCS (Academic Research and Careers for Students)

Reviewer

2023 IEEE/ACM Transactions on Networking (ToN))

Conference Session Chair

2023, 2024 COMSNETS

Shadow Program Committee member

2020 The ACM SIGOPS European Conference on Computer Systems(EuroSys)

Invited talks and tutorials (recent)

June 2025 Networked Systems Research @ PNL-IIITD
Invited faculty talk at IndoSys 2025

Jan 2025 Evolution of Ethernet Transport for AI and HPC Networking
Invited keynote at the MLNetSec Workshop colocated with ICDCN 2025, January 7, 2025

July 2024 Workload-aware in-network crypto scheduler for FPGA NICs
Seminar talk at the Dagstuhl Seminar 24291 on "Programmable Host Networking", Schloss Dagstuhl, July 2024

Dec 2023 Linux Networking Basics
Tutorial conducted at ACM India Winter School on Full-stack Networking, IIIT-Delhi, December 18, 2023

July 2023 Transitioning from student to a Research career
Invited Panelist for panel discussion at Maitreyee 2023, IBM Research - India, July 27, 2023

July 2023 P4 for All
Invited tutorial at Workshop on "SDN: Software Defined Networking Architectures and Applications" at IIIT Kottayam, July 17-23, 2023. Sponsored by SERB, Accelerate Vigyan, Karyashala.

May 2022 Building programmable networks
Invited talk at ACM SIGMETRICS PerfNA 2022.

Jan 2022 P4 Language and Programable Data Planes
Tutorial conducted at COMSNETS 2022.

July 2020 Leveraging Programmable Networks to Accelerate Networked Systems
University: IIT Hyderabad (January 2021), IIIT Delhi (December 2020), BITS Pilani (Goa) (July 2020)

Current students

Ph.D. students

1 Sumit Kumar –> Oct'23 – present
2 Aditya Peer –> July'25 – present
3 Palak Mishra –> July'25 – present
4 Rajat Bhardwaj –> July'26 – present
5 Mohammad Ali –> July'26 – present
6 Shivansh Bhatnagar –> July'26 – present

MTech thesis/capstone/dual-degree students

1 Prakhar Agarwal –> Jan'26 – May'27
2 Harsha Verma –> Jan'26 – Dec'26
3 Aditya Malik –> Jan'26 – Dec'26
4 Aditya Sai Teja –> Jan'26 – Dec'26
5 Amisha Patel –> Jan'26 – Dec'26

Teaching (recent)

Core courses

2021–2023 Computer Networks (CSE232)
2025–2026 Graduate Systems (CSE638)
2026 Operating Systems (CSE231)

Elective courses

2021–2023 Programmable Networking (CSE567))
2025–2026 Networks for AI/ML Systems (CSE587))

New courses designed @ IIITD

2022 Programmable Networking (CSE567)
2024 Introduction to Graduate Systems (Core course for New MTech CSE Curriculum 2024)
2025 Networks for AI/ML Systems (CSE587)

Collaborators

Academic collaborators

IITH Praveen Tammana
BITS Goa Vinayak Naik
IIIT-Delhi Sambuddho Chakravarty
IIIT-Delhi Sumit Darak

Industry collaborators

Marvell Tech Abed Mohammad Kamaluddin, Satananda Burla
IBM research Seep Goel, Priyanka Naik, Chander Govindarajan, Kavya Govindarajan

Achievements and Awards (recent)

2025 My Ph.D. student, Aditya, received the **Best M.Tech. thesis award (CSE)** @ The 14th Convocation, IIIT-Delhi. Thesis title: Workload-aware In-network Crypto Primitives for FPGA NICs

2025 Received **Distinguished Teaching Excellence Award** for AY 2024-25.

2024 My Ph.D. student, Neeraj, received prestiguous **TCS research fellowship (TCS RSP)**.

2024 Received the invitation to the prestigious **Dagstuhl Seminar on Programmable Host Networking**

2024 Selected to participate in **ACM India Pingala Interactions in Computing (PIC'24)** – Selected 50 candidates from India/International, Nobody else from IIITD

2023, 2024 Received **Teaching Excellence Award** for AY 2022-23 and AY 2023-24

2023 My MTech student, Shaguftha Kottur, received **best MTech thesis award** in the CSE department, IIITD. Thesis title: Implementing ChaCha Based Crypto Primitives on Programmable SmartNICs

2023 ACM India accepted our application to **host ACM India Winter school** on Full-stack Networking and awarded funding of $\sim$ 6 lakh rupees. School co-ordinators: Rinku Shah, Arani Bhattacharya, Priyanka Naik

2023 My BTech project students, Vedant Bothra and Aditya Peer, received **Summer Undergraduate Research Fellowship (SURF) Excellence award**. Project title: Realising AQM Algorithms on Programmable Network Switch Hardware